SLIDE-QUIZ EXERCISE · ATMOS01

Atmospheric Pressure

Understanding the Pressure of the Air Around Us

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EXERCISE OVERVIEW

Atmospheric Pressure

This exercise explores the weight of the air we live beneath. Four content pages move from the idea of pressure itself to the way it was first measured and the many units used to report it today. The first page defines pressure and atmospheric pressure and explains why the air presses on us at all; the second follows Torricelli's mercury barometer to the modern digital sensor; and the last two sort out the units, from the Pascal and the millibar of the weather map to the psi of a car tire, the mm Hg of a blood-pressure reading, and the in Hg of an aircraft altimeter. Every content page has a narrated instruction panel you can play, two pages have interactive figures, and the graded questions number six in all.

The Exercise, Page by Page

Page	What you'll do
Introduction	A short video explains what to expect on the pages that follow.
1. What Is Pressure?	Pressure as force divided by area, and atmospheric pressure as the weight of the air above you. Two questions.
2. Measuring Atmospheric Pressure	Torricelli's mercury barometer, the barometer simulation, and modern barometers. Two questions.
3. Pressure Units — Part 1	Pascal, Atmosphere, bar, millibar, and hectopascal; Blaise Pascal. One question.
4. Pressure Units — Part 2	psi, mm Hg (Torr), and in Hg; blood pressure and the aircraft altimeter. One question.
Conclusion	A closing video, your score, and the Submit Quiz button.

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TRY IT IN THE EXERCISE

On each page, press Start to hear the narration. On page 2, drag the barometer's pressure slider and watch the mercury column, the unit readouts, and the forecast all change together; on page 4, study the altimeter diagram. Keep the Numbers to Know table at hand as you go: most of the graded questions turn on a definition, a name, or a unit.

Learning Objectives

After this exercise you should be able to:

- Define pressure as force divided by area, and state its metric (SI) unit, the Pascal.
- Define atmospheric pressure, and explain why it decreases with increasing altitude.
- Describe Torricelli's mercury barometer and explain what holds the mercury column up.
- State the general relationship between atmospheric pressure and the weather.
- Name the modern instruments that have replaced the mercury barometer.
- Give the value of one Atmosphere in Pa, bar, mbar, hPa, psi, mm Hg, and in Hg, and convert between them.
- Identify where each unit is used: weather maps, tires, medicine, and aviation.

QUICK REFERENCE

Key Terms

Term	Definition
Pressure	The force applied perpendicularly to a surface divided by the area of that surface. Pressure is a scalar quantity.
Atmospheric pressure	The weight of the column of air directly above a given altitude, divided by the cross-sectional area of that column.
Pascal (Pa)	The metric (SI) unit of pressure: one Newton of force spread over one square meter. A very small amount of pressure.
Barometer	An instrument that measures atmospheric pressure.
Mercury barometer	Torricelli's 1643 instrument: a glass tube of mercury inverted into a dish of mercury. The air pressing on the dish holds up a column about 760 mm tall at sea level.
Torricellian vacuum	The empty space above the mercury at the top of the tube — the first sustained man-made vacuum.
Aneroid barometer	A mechanical barometer with a sealed, flexible metal capsule that expands and contracts as the pressure changes (Lucien Vidie, 1844).
Digital (MEMS) barometer	An electronic barometer whose tiny pressure-sensing chip changes its electrical properties as the pressure changes; found in phones, weather stations, and aircraft.

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Term	Definition
Barograph	A recording barometer that traces pressure continuously on paper wrapped around a rotating drum.
Atmosphere (Atm)	A non-SI unit equal to the average air pressure at sea level: 1 Atm = 101,325 Pa.
Bar / millibar (mbar, mb)	1 bar = 100,000 Pa; 1 mbar = 1/1000 bar = 100 Pa. The millibar was the standard weather unit for most of the 20th century.
Hectopascal (hPa)	100 Pa. Numerically identical to the millibar; the unit preferred by the World Meteorological Organization.
Pascal's Principle	An external pressure applied to a liquid is extended, undiminished, to every point in the liquid and to the walls of its container.
psi	Pounds per square inch: one pound of force on one square inch. Used for tire, scuba-tank, and industrial pressures in the US.
mm Hg / Torr	The pressure that supports a 1 mm column of mercury. The Torr (for Torricelli) is 1/760 of an Atmosphere. Used for blood pressure.
in Hg	Inches of mercury. Used for US weather reports and for setting aircraft altimeters.
Altimeter	An aircraft instrument that reads altitude from the outside (static) air pressure, which falls as the aircraft climbs.
Kollsman window	The small window on an altimeter where the pilot dials in the local barometric pressure (in in Hg in the US).

QUICK REFERENCE

The Units at a Glance

Unit	Symbol	1 Atm equals	Where it's used
Pascal	Pa	101,325 Pa	The metric (SI) unit; science
Atmosphere	Atm	1 Atm (the reference)	Comparing any pressure with sea level
Bar	bar	1.01325 bar	Oceanography, meteorology, industry
Millibar	mbar, mb	1,013.25 mbar	Weather maps, aviation weather
Hectopascal	hPa	1,013.25 hPa	Modern weather reports (WMO standard)
Pounds per square inch	psi	14.696 psi	Tires, scuba tanks, hydraulics (US)
Millimeters of mercury	mm Hg, Torr	760 mm Hg = 760 Torr	Blood pressure, vacuum work
Inches of mercury	in Hg	29.921 in Hg	US weather reports, altimeter settings

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QUICK REFERENCE

Numbers to Know

Value	What it refers to
101,325 Pa	One Atmosphere: average air pressure at sea level
14.7 psi	The same pressure in pounds per square inch (14.696 precisely)
760 mm (29.92 in)	Height of Torricelli's mercury column at sea level
1 Pa	One Newton of force on one square meter: *1 Pa = 1 N / 1 m²*
100,000 Pa	One bar
100 Pa	One millibar, which equals one hectopascal
1.3%	How much greater one Atmosphere is than one bar
1643	Torricelli invents the mercury barometer
1648	Puy de Dôme experiment shows pressure falls with altitude
1844	Lucien Vidie's aneroid barometer
870–900 mbar	Pressure near the eye of a very powerful hurricane
30–35 psi	Typical car tire pressure
1,020 mb and up	"Fair" on the barometer simulation; below 1,000 mb it reads Rain, and below 980 mb storms

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EXERCISE CONTENT

Page-by-Page Summary

Part 1 — What Is Pressure?

Pressure is the force applied perpendicularly to a surface divided by the area of that surface, $P = F / A$. It is a scalar: it has a size but no direction. Pressure depends on both the force and the area it acts on, so a small force on a tiny area makes a very large pressure — a needle pierces skin where a broad palm cannot. The metric (SI) unit is the Pascal, $1\ Pa = 1\ N / 1\ m^2$, a very small amount of pressure.

Atmospheric pressure is the weight of the column of air directly above a given altitude, divided by the area of that column. We live at the bottom of an ocean of air; the higher you go, the less air is above you and the lower the pressure. At sea level it is 101,325 Pa, or 14.7 pounds on every square inch. We don't feel it because the air inside our bodies, in the lungs, pushes outward just as hard.

Part 2 — Measuring Atmospheric Pressure

In 1643 Evangelista Torricelli (1608–1647), a student and successor of Galileo, filled a glass tube with mercury and turned it upside down into a dish of mercury. The column settled at about 760 mm, or 29.92 inches. A vacuum does not "pull" it up: the weight of the atmosphere pressing on the open dish holds the column up, so its height is a direct measure of the air pressure. The empty space at the top, the Torricellian vacuum, was the first sustained man-made vacuum. "We live submerged at the bottom of an ocean of air," he wrote.

The barometer simulation shows the column rising and falling with the pressure, and the forecast moving from Fair at high pressure to Rain and storms at low pressure: the higher the pressure, the fairer the weather.

Because mercury is extremely toxic, it has been replaced by the aneroid barometer (a sealed, flexible metal capsule, 1844), the electronic or digital barometer (a tiny MEMS pressure chip, found in smartphones, aircraft, and weather stations), and the barograph, which records pressure on a rotating paper drum.

Part 3 — Pressure Units, Part 1

Every unit describes the same quantity, measured against one reference: 1 Atm = 101,325 Pa. The Pascal, named for Blaise Pascal, is so small that larger units are often handier. The bar (1909) is exactly 100,000 Pa, so 1 Atm = 1.01325 bar, only 1.3% more. The millibar is a thousandth of a bar, or 100 Pa, so 1 Atm = 1,013.25 mbar. The hectopascal is also 100 Pa — numerically identical to the millibar — and is the World Meteorological Organization's preferred unit.

The units are linear, so converting is just multiplying: $0.5\ Atm = 50{,}662.5\ Pa = 506.625\ mbar$.

Blaise Pascal (1623–1662) showed with the 1648 Puy de Dôme experiment that air pressure falls with altitude, built an early mechanical calculator, and gave his name to Pascal's

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Principle: a pressure applied to a liquid is passed, undiminished, to every point in the liquid and to the walls of its container.

Part 4 — Pressure Units, Part 2

Three units are especially common in the United States. Pounds per square inch, with 1 Atm = 14.696 psi, is used for tires (a typical car runs 30–35 psi), scuba tanks, and hydraulics. Millimeters of mercury come straight from Torricelli's barometer: 1 Atm = 760 mm Hg = 760 Torr, and blood pressure is always reported in mm Hg. Inches of mercury, with 1 Atm = 29.921 in Hg, are used in US weather reports and to set aircraft altimeters. Again: *0.5 Atm = 7.35 psi = 380 mm Hg = 14.96 in Hg*.

The altimeter diagram shows how a plane knows its height. Its static port senses only the outside air pressure; as the plane climbs, that pressure falls and the altimeter needle turns. The pilot dials the local pressure into the Kollsman window so the altimeter reads the correct altitude.

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — What Is Pressure?

1. Define pressure in words, and write the equation for it.

2. What is the metric (SI) unit of pressure, and how is it defined? Is it a large or a small amount of pressure?

3. Using $P = F / A$, explain why a sharp needle can pierce skin while a broad palm pushing with the same force cannot.

4. Define atmospheric pressure.

5. How does atmospheric pressure change as you go higher, and why?

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6. At sea level the atmosphere pushes on every square inch of your body with about 14.7 pounds of force. Why don't you feel it?

Part 2 — Measuring Atmospheric Pressure

1. Who invented the mercury barometer, and in what year? What was his connection to Galileo?

2. Describe Torricelli's experiment. At what height did the mercury column settle at sea level, in millimeters and in inches?

3. What holds the mercury column up in the tube? Is the vacuum at the top "pulling" it up?

4. What is the empty space above the mercury called, and why was it historically important?

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5. In the barometer simulation, what happens to the mercury column and to the forecast when you raise and lower the pressure? State the general rule linking pressure and weather.

6. Why are mercury barometers no longer used? Name the three modern instruments on this page and say briefly how each works.

Part 3 — Pressure Units, Part 1

1. What is one Atmosphere (Atm), and what is its value in Pascals?

2. Give the value of 1 Atm in bar, in millibars, and in hectopascals.

3. How many Pascals are in one millibar and in one hectopascal? Why was switching weather maps from millibars to hectopascals so easy?

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4. All pressure units are linear. Convert 0.5 Atm into Pascals and into millibars. Show your work.

5. Who was Blaise Pascal? What did the 1648 Puy de Dôme experiment show, and what does Pascal's Principle say?

6. Near the eye of a very powerful hurricane the pressure can drop to about 900 mbar. Roughly what fraction of one Atmosphere is that?

Part 4 — Pressure Units, Part 2

1. What does psi stand for? What is 1 Atm in psi, and where is psi used?

2. What is a millimeter of mercury, and how is it connected to Torricelli's barometer? How does it compare to the Torr?

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3. In what unit is blood pressure reported?

4. What is 1 Atm in inches of mercury, and where is in Hg the standard unit?

5. How does an aircraft altimeter use air pressure to show altitude? What is the Kollsman window for?

6. Convert 0.5 Atm into psi, mm Hg, and in Hg.

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YOUR NOTES

Notes

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